



Al-Balqa Applied University
Faculty of Engineering Technology, Scientific Basic Science Department, MID Exam

Linear Algebra.

Question Number	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	A	B	C	C	D	A	B	C	D	B

Question One: The system $x + y = \pi$, $2x + y = 2\pi$ has

- A) a unique solution B) infinitely many solutions C) no solution
D) two solutions E) None

Question Two: If the system $x + 2y = a$, $3x + 6y = a^2$ has infinitely many solutions, then the value of the positive constant a equals

- A) 2 B) 3 C) 4 D) 5 E) None

Question Three: For a nonsingular matrix A with

$$A^{-1} = \begin{bmatrix} 1 & 1 \\ 0.5 & 1.5 \end{bmatrix}$$

Then the solution of the system $A^{-1}X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is

- A) $(-3, -2)$ B) $(2, -2)$ C) $(-2, 2)$ D) $(-2, 3)$
E) None

Question Four: Let I be the 3×3 identity matrix, then

$$((\text{trace}(I) I)^{-1})^t =$$

- A) $\frac{1}{5} I$ B) $\frac{1}{4} I$ C) $\frac{1}{3} I$ D) $\frac{2}{9} I$ E) None

Question Five: Consider a $n \times n$ matrix A with $\det(A) = 2$. If $\det(nA^{-1}) = \frac{27}{2}$, then $n =$

- A) 5 B) 2 C) 4 D) 3 E) None

Question Six: Consider the vector space $V = \mathbb{R}^3$ where

$$(x_1, y_1, z_1) \oplus (x_2, y_2, z_2) = (x_1 + x_2 - 1, y_1 + y_2, z_1 + z_2 - 1)$$

Then the zero vector $OV =$

- A) (1,0,1) B) (1,1,0) C) (1,0,0)
D) (0,0,0) E) None

Question Seven: The constants that make $p = x^2 - 6x$ a linear combinations of $p_1 = x^2$ and $p_2 = 3x$ are

- A) 1, -3 B) 1, -2 C) 1, 3 D) 1, 2 E) None

Question Eight: For the matrix

$$A = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}$$

We have $\det(A^{-1}) + \text{trace}(A^{-1}) =$

- A) 3 B) 2 C) 14 D) 15 E) None

Question Nine: If the matrix

$$A = \begin{bmatrix} 0.5 & a - 4 & 0 \\ 1 & 0.5 & 1 \\ 0 & 1 & 0.5 \end{bmatrix}$$

is symmetric, find the value of the constant a

- A) 2 B) 3 C) 4 D) 5 E) None

Question Ten: If $\det \begin{bmatrix} a & b \\ 1 & 2 \end{bmatrix} = 5$; then $\det \begin{bmatrix} 2 & 4 \\ -a & -b \end{bmatrix} =$

- A) 20 B) 10 C) -10 D) 5 E) None

Question Eleven: Find : if any , the solution(s) of the system
 $x+y=3$, $2x+y=5$ using Gauss-Jordan Elimination

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 5 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -1 & -1 \end{array} \right]$$

$$R_2 = -R_2 \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array} \right]$$

$$R_1 = R_1 - R_2 \quad \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

Solution: $(x, y) = (2, 1)$

Question Twelve: Show that $W = \{(a, b) \in \mathbb{R}^2 : a\}$ is the system
 $\mathbb{R}^2$ under the natural vector addition and scalar multiplication

Proof:

1. Check Closure under addition: Let $u=(a, b)$, $v=(c, d)$ be in W .
 We need $u+v=(a+c, b+d)$ in W .

Since u, v are in W , $a=b+1$ and $c=d+1$.

Check if $u+v$ satisfies the condition: $a+c = (b+1) + (d+1) = b+d+2$.

For $u+v$ to be in W , it must satisfy $a+c = (b+d)+1$.

Since $b+d+2 \neq b+d+1$, W is not closed under vector addition.

Thus, W is not a subspace of $\mathbb{R}^2$. (Counterexample: $(1, 0)$ and $(2, 1)$ are in W , but their sum $(3, 1)$ is not, since $3 \neq 1+1$).